

Project Title: PetCare Manager
A Pet-Sitting Business Management System
(Boarding, Walking, Day Care, House Sitting)

Team 11

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Project Description

PetCare Manager is a web platform that lets independent pet-sitting businesses manage bookings, clients, and pets directly, without relying on large marketplace apps. Sitters currently rely on platforms like Rover or Wag to find clients, but those platforms typically charge around **20-30% commission** on every booking. Many sitters build a base of repeat clients over time, yet continue paying that commission indefinitely because they have no independent system to manage scheduling, client records, pet care details, and payments on their own. PetCare Manager is **motivated** by this ongoing cost and offers sitters a self-hosted alternative for the client relationships they've already built.

The **primary goal** of PetCare Manager is to let a pet-sitting business track every client, pet, booking, and payment in one dashboard, while eliminating the recurring commission fees charged by third-party platforms. The system stores detailed pet profiles (species, breed, feeding instructions, medications, behavioral notes, photos) so that any sitter assigned to a booking has the information they need to safely care for the pet. Bookings can span multiple pets and multiple services (for example, boarding two dogs while also walking a third), and each booking tracks its own payment and review.

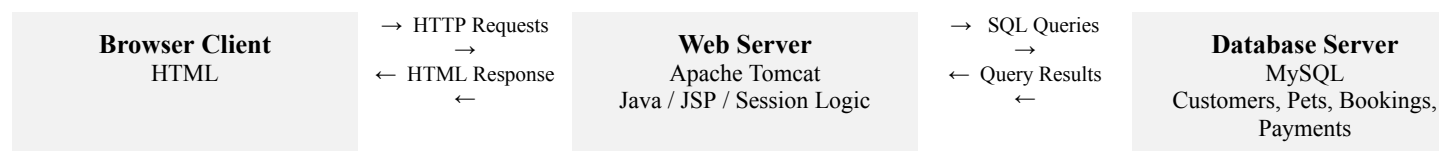
The **major stakeholders** include pet owners (clients), who need a simple way to book trusted care and share instructions about their pets, and the sitter (the business owner), who needs an organized way to manage schedules, client information, and payments. Since this is a solo pet-sitting business, the sitter and the admin are the same person — one account that manages services, bookings, and payments rather than separate sitter/admin roles.

Our specific **innovation** is that PetCare Manager belongs to the sitting business rather than to a marketplace intermediary. Unlike existing platforms that take a percentage of every transaction in exchange for client discovery, PetCare Manager is built for sitters who already have a client base and simply need booking, scheduling, and record-keeping tools — letting them keep the full amount clients pay for their services.

System Environment

System Structure

PetCare Manager follows a three-tier architecture:



Client (Browser) Layer: Clients and sitters access the system through a web browser. The interface is built using HTML.

Application (Web) Layer: The application logic runs on Apache Tomcat using Java and JSP. This layer handles login, session management, permission checks, booking logic, and price/availability calculations.

Database Layer: MySQL stores all persistent data, including customers, pets, addresses, services, bookings, payments, care instructions, and reviews.

Hardware and Software Used

- OS: Windows / macOS / Linux
- Web Server: Apache Tomcat
- Database: MySQL Community Server
- Languages: Java, JSP, SQL, HTML

Database Design

The system is organized around the following core tables:

#	Table	Purpose
1	Customer	Pet owner information
2	Pet	Information about each pet
3	Address	Customer or service location
4	Service	Walking, sitting, boarding, day care, etc.
5	Booking	Scheduled pet-sitting appointments
6	Booking_Pet	Connects bookings with pets
7	Booking_Service	Connects bookings with services
8	Payment	Payment information and status
9	Care_Instruction	Feeding, medication, and special instructions
10	Review	Customer ratings and comments

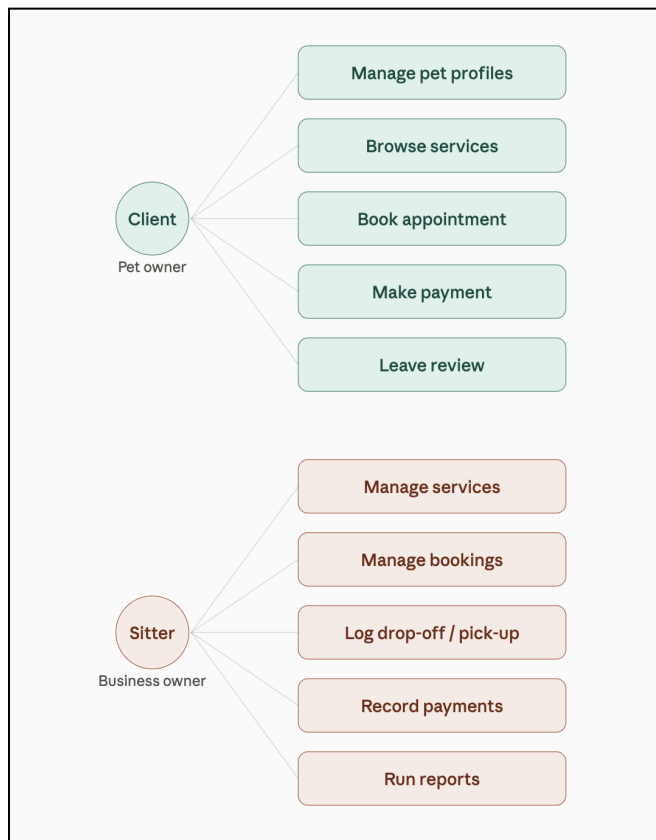
Pet Profile Fields

Each Pet record stores the details a sitter needs to safely care for the animal:

- Name, species, breed, age, and gender
- Color and physical markings
- Favorite toys
- Feeding instructions
- Medication needs
- Behavioral concerns
- Emergency notes
- Photos

Functional Requirements

PetCare Manager offers different functions depending on the user's role: Client (pet owner) or Sitter (the business owner). Clients mainly use the system to manage their pets' profiles, book services, and pay for completed visits. The Sitter account manages the full business: the service catalog, all bookings, pet care instructions, payments, and client accounts.



Client (Pet Owner) View

User Authentication

- Clients must log in using their email and password before they can book services or add pets. There will also be a registration portal if they are a new user.
- This prevents fake bookings and keeps accounts accountable.
- Passwords are stored securely using hashing.
- After login, the system creates a secure session for the client.
- Visitors who are not logged in can still view the service directory and pricing, but they can't book a service or add a pet.

Pet Profile Management

- Clients can add one or more pets to their account, each with:
 - Name, species, breed, age, and gender
 - Color and physical markings
 - Favorite toys
 - Feeding instructions and medication needs
 - Behavioral concerns and emergency notes
 - Photos
- Pet profiles can be edited at any time and are attached to future bookings automatically.

Service Directory

- Clients can view a list of services offered by the business (e.g., "Dog Walking", "Overnight Boarding", "Day Care", "House Sitting").
- Each service shows:
 - Service name and category
 - Price and duration
 - Current availability
 - Assigned sitter(s), if applicable

Booking

- Clients can request a booking by selecting one or more pets, one or more services, and a date/time range.
- The system records:
 - Customer ID and Pet ID(s) via Booking_Pet
 - Service ID(s) via Booking_Service
 - Requested start/end date and time
 - Booking status (requested, confirmed, in progress, completed, cancelled)
- The system prevents double-booking a sitter for overlapping time slots.
- Each booking automatically pulls in the relevant Care_Instruction records for the pets involved, so the assigned sitter has what they need.
- The booking status is possible to be changed by the sitters themselves. The client will also have the option of cancelling the booking within a specific set time range.

Care Instructions

- Clients can attach or update special instructions per pet per booking, such as feeding times, medication dosages, or behavioral notes for that specific stay.

Payment

- Once a booking is confirmed, the client can view the invoice showing the amount due.

- Clients can pay using cash, credit card, Venmo, or Zelle — however the sitter and client arrange it.
- The system records payment amount, method used, status (pending, paid, refunded), and timestamp in the Payment table, so every booking has an accurate record regardless of how it was paid.
- Clients can view their full payment history.

Reviews

- After a booking is marked completed, the client can leave a rating and comment for the sitter, stored in the Review table.
- Clients will also have an option of leaving a tip for the sitter if they would like.

Booking History

- Clients can view all upcoming and past bookings along with their current status.

Sitter (Business Owner) View

Since this is a solo pet-sitting business, one account — the Sitter — handles everything an owner/operator needs: managing the service catalog, running bookings day-to-day, and keeping the books straight.

Sitter Login

- The sitter logs in with a single account that has full access to the business's data.

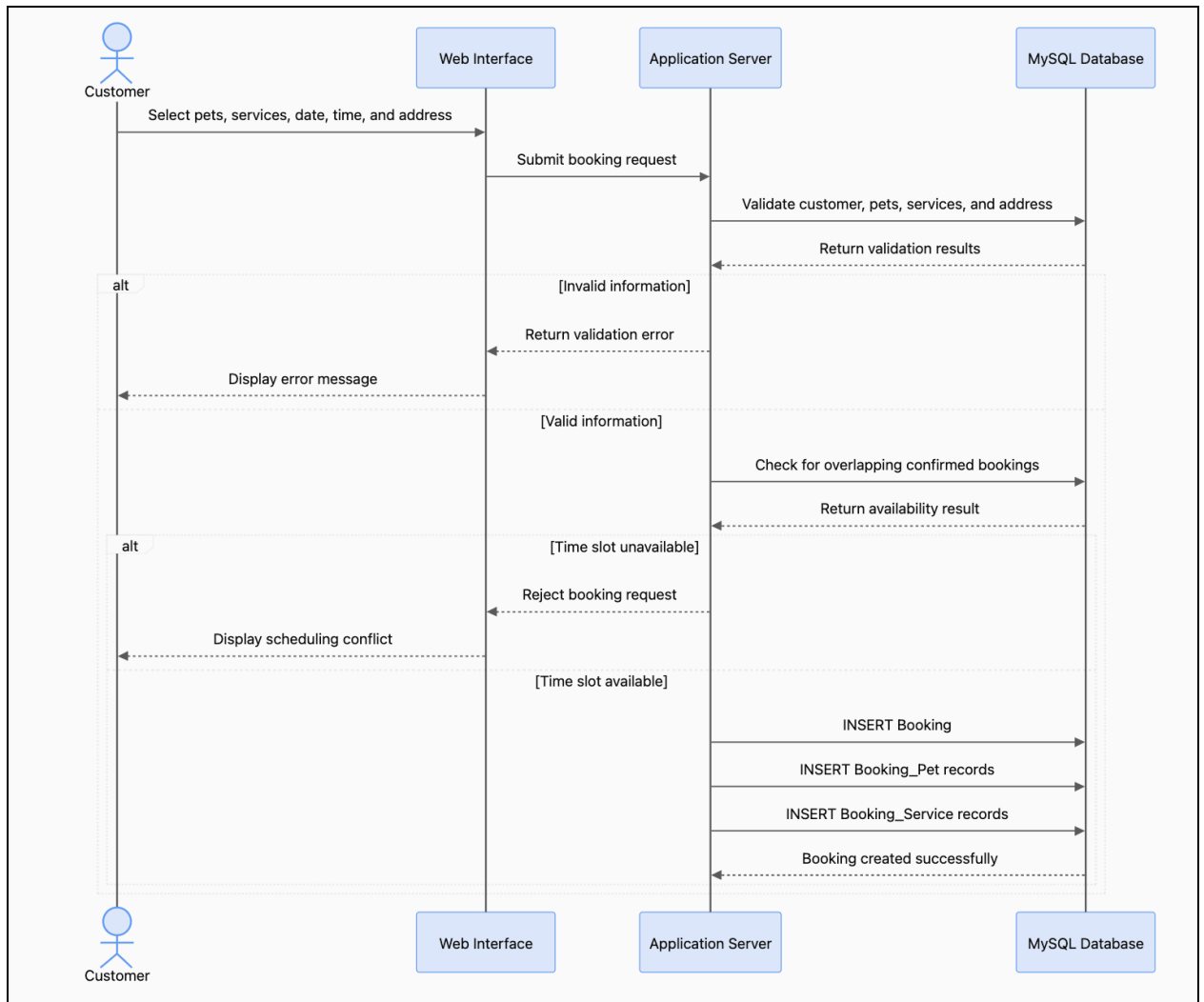
Manage Services

- The sitter can add, edit, or deactivate the services offered (Walking, Boarding, Day Care, House Sitting), including:
 - Pricing and duration
 - Availability / hours
 - Capacity (e.g., maximum pets per day)

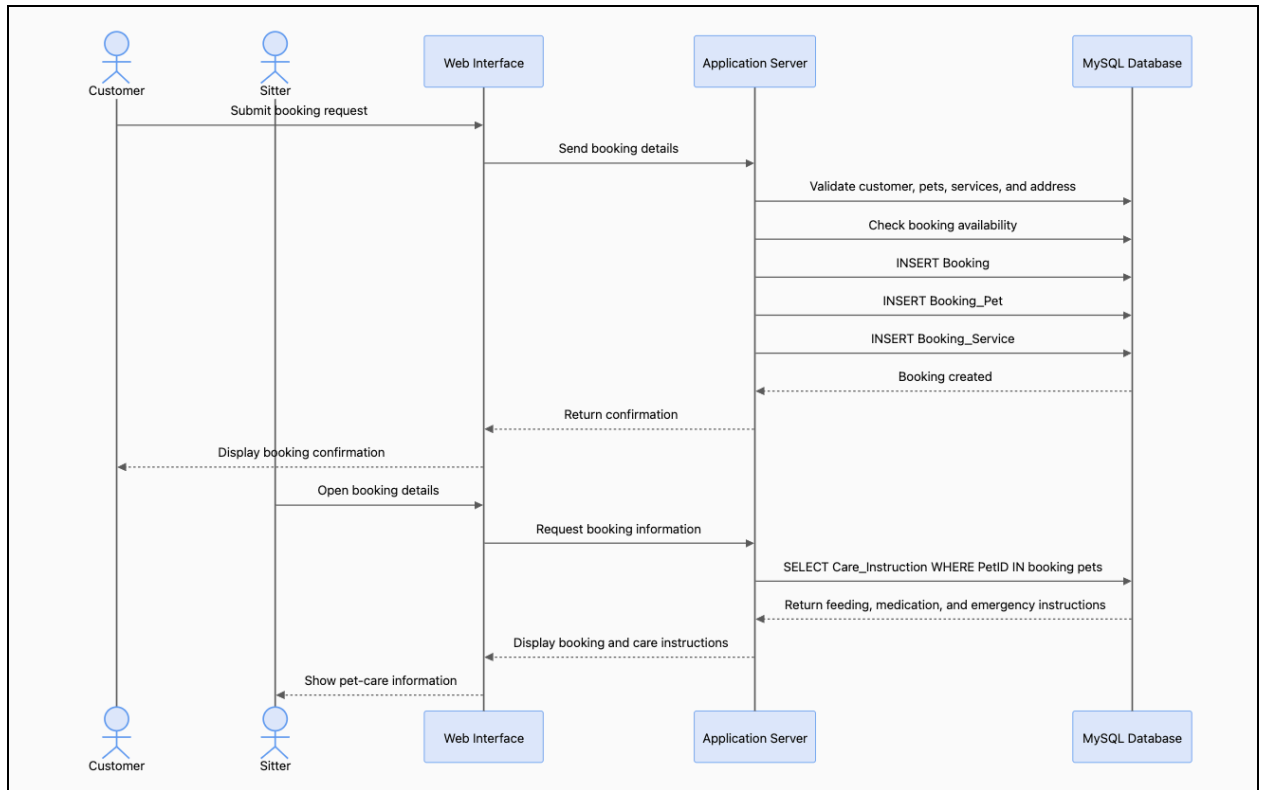
Booking Management

- The sitter can view every booking, including client contact info, pet profiles, and care instructions.
- The sitter can accept, decline, or reschedule a booking request.

Example :



Sequence diagram for submitting a pet-care booking request.



After a booking is successfully created, the customer receives a confirmation. When the sitter later opens the booking, the system retrieves the related pets and displays their care instructions.

Drop-off / Pick-up Events

- For boarding and day care, the sitter can log a drop-off event when the pet arrives and a pick-up event when the pet is collected.
- The system records the timestamp for each event and calculates the actual duration of care, which is used to confirm the final invoice amount.

Payments

- The sitter marks each booking's payment as received (cash, credit card, Venmo, or Zelle) and can view a full payment/earnings history across all bookings.

Client Moderation

- The sitter can suspend a client account that abuses the system (e.g., repeated no-shows or fake bookings) and review flagged bookings or reviews.

Reporting

- The sitter can run reports on revenue, booking volume, and most-booked services to track how the business is doing over time.

Non-Functional Issues

Graphical User Interface

- The interface must be easy to use and quick to navigate on both desktop and mobile, since sitters are often on the go between appointments.
- Clients should be able to book a service in just a few clicks.
- The GUI should include:
 - A calendar view of bookings
 - Color-coded booking status (requested, confirmed, in progress, completed)
 - Clear pet profile cards with photos
 - A clean dashboard for sitters and admins

Security

- Passwords are stored using secure hashing.
- Sessions expire after inactivity.
- All input fields are validated to prevent SQL injection.
- Because payments include cash, Venmo, and Zelle (methods the platform doesn't process directly), the system never stores card numbers or bank details — it only records the amount, method, and status of each payment as entered by the sitter.
- Sitter actions (edits, cancellations, refund status changes) are recorded in audit logs.
- Permission checks are enforced on the server side.

Access Control

- The system uses role-based access control:
 - Clients: can manage only their own pets, bookings, and payments
 - Sitter: has full control over services, bookings, payments, and client moderation

Performance

- The system must handle peak booking periods (e.g., holiday boarding requests) without slowing down.
- Database indexing will be used on key fields like sitter ID, booking date, and service ID to improve query speed.

Reliability and Data Integrity

- The system must prevent:
 - Double-booking a sitter for overlapping time slots
 - Orphaned Booking_Pet or Booking_Service records
 - Incorrect payment status transitions (e.g., marking a booking paid without a completed payment record)

Audit and Accountability

- All sitter and admin actions (edits, cancellations, account suspensions) are recorded in audit logs to support dispute resolution.